

Question Paper

Exam Date & Time: 04-Jan-2026 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026
Advanced Software Engineering [IT_4417]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze how component-based software development enhances software reuse and scalability. (5)
A) Illustrate your answer with an example of a layered software architecture used in enterprise applications
Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Compare the Agile process model with the traditional Waterfall approach in terms of flexibility, iteration, and customer feedback. Discuss how Agile supports rapid change management. (5)
Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Examine how organizational structures and governance policies influence the implementation of security and dependability within large-scale software systems. Support your discussion with relevant organizational models. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- B) Break down the concept of sociotechnical resilience and analyze its implications for the design of fault-tolerant systems, considering both human and technical subsystems. (5)
Course Outcome: "CO3" / Cognitive Level: "Evaluate"
- 3) Analyze the role of security testing in the software assurance process. How do strategies like penetration testing uncover vulnerabilities? (5)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Explain the Component-Based Software Engineering process model. Compare it with traditional software development life cycles in terms of reuse and integration. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Evaluate the role of Component-Based Software Engineering (CBSE) in agile development environments. Is CBSE compatible with iterative delivery and continuous refactoring principles? Justify with suitable examples. (5)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Analyze the architectural layers of Service-Oriented Architecture (SOA) and their role in service orchestration and interoperability within distributed enterprise systems. (5)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 5) Explain the effectiveness of system operation and evolution strategies in maintaining long-lived software systems. Evaluate how re-engineering, refactoring, and maintenance ensure architectural stability. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- B) Compare architectural patterns used in real-time systems. How do these patterns address timing, concurrency, and synchronization constraints using AI (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4417:Advanced Software Engineering

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	20
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4417:Advanced Software Engineering

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	2	10(20%)
4	Analysis	4	20(40%)
5	Create	4	20(40%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)